THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS TWO-CHAPTER TEST SET D

Chapters 9 & 10: **Propositions and their Converses, How Quantities Combine: Understanding Data** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A number that is the product of two different primes has exactly: 1

(a) 2 factors **(b)** 6 factors **(c)** 4 factors **(d)** 3 factors

2. A counterexample to a proposition is an instance that: 1

(a) proves the proposition
(b) satisfies both parts
(c) satisfies neither part
(d) satisfies the 'if' part but fails the 'then' part

3. Fermat claimed that every number of the form $2^{2^n}+1$ is prime. Euler disproved this by showing that for $n=5$ the number is: 1

(a) prime **(b)** negative **(c)** a perfect square **(d)** composite

4. "If $a=b$, then $a+5=b+5$." This proposition and its converse are: 1

(a) both true **(b)** both false
(c) only the proposition is true **(d)** only the converse is true

5. **Assertion (A):** The number 12 is a counterexample to "If a number is a multiple of 3, then it is a multiple of 9". 1

Reason (R): 12 is a multiple of 3 but not a multiple of 9.

(a) Both A and R are true and R is the correct explanation of A
(b) Both A and R are true but R is not the correct explanation of A
(c) A is true but R is false
(d) A is false but R is true

6. Marks of 50, 80, 65 are combined with weights 5 : 3 : 2. The weighted average is: 1

(a) 63 **(b)** 61 **(c)** 64 **(d)** 62

7. How much water must be added to 12 litres of a 10% solution to reduce its concentration to 8%? 1

(a) 3 litres **(b)** 5 litres **(c)** 6 litres **(d)** 4 litres

8. The weighted mean of values $x_1, x_2, \ldots, x_n$ with weights $w_1, w_2, \ldots, w_n$ is: 1

(a) $\dfrac{w_1+w_2+\cdots+w_n}{x_1+x_2+\cdots+x_n}$ **(b)** $w_1x_1 \cdot w_2x_2 \cdots w_nx_n$ **(c)** $\dfrac{x_1+x_2+\cdots+x_n}{n}$ **(d)** $\dfrac{w_1x_1+w_2x_2+\cdots+w_nx_n}{w_1+w_2+\cdots+w_n}$

9. A house used 150, 450, 250 and 150 units of electricity on lighting, cooling, kitchen and other appliances in 2018. In a 100% stacked bar for 2018, the length of the cooling segment is: 1

(a) 40% **(b)** 45% **(c)** 50% **(d)** 90%

10. **Assertion (A):** If equal quantities of 10% and 20% jaggery lemonade are mixed, the result has 15% jaggery. 1

Reason (R): With equal quantities, the weighted average becomes the simple average of 10% and 20%.

(a) Both A and R are true and R is the correct explanation of A

(b) Both A and R are true but R is not the correct explanation of A

(c) A is true but R is false

(d) A is false but R is true

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Find a counterexample to "If n is an even number, then $2^n + 1$ is a prime number". 2

12. Explain with an example why finding one counterexample is enough to disprove a proposition. 2

13. What does a 'weight' mean in a weighted average? 2

14. A 100% stacked bar shows Shop P: 60% sales in the morning; Shop Q: 40% in the morning. Can we say P sold more in the morning? Give a reason. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A teacher says "If a number is a multiple of 8, it is a multiple of 2 and of 4." Write the converse. Does 'divisible by 2 and 4' suffice to test divisibility by 8? Give an example. 3

16. State two situations from real life where the weighted mean is used and say what the weights are. 3

17. Write the converse of "If the sum of the digits of a number is a multiple of 3, then the number is divisible by 3." Is it true? 3

OR

Show with an example that "If two triangles have equal areas, then they are congruent" is false, and write its converse and its truth. 3

18. A bar chart shows two bars: Family A spends ₹7000 with 20% on healthcare; Family B spends ₹4000 with 30% on healthcare. Which family spends more rupees on healthcare? 3

OR

Show that mixing equal quantities of two solutions with concentrations $p\%$ and $q\%$ gives $\frac{p+q}{2}\%$. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. (i) Why does the argument "the number of factors is odd, so a factor pair repeats" prove only the converse of 'perfect squares have an odd number of factors'? 5

(ii) How can the argument be extended to prove the proposition itself?

OR

A trader says: "If I sell more articles, my profit is more." 5

(i) Write the converse.

(ii) A shop sells 10 pens at ₹4 profit each and then 20 pens at ₹1 profit each; show that selling more gave less profit.

(iii) Give a situation to show that the converse can be false.

(iv) What are such examples called?

20. (i) Find the mean depth of a pit (Brahmagupta's rule) if the depth is 2 m for a length of 3 m, 5 m for 4 m and 3 m for 3 m. 5

(ii) Explain why a simple average of the depths would be misleading.

OR

(i) The average daily rainfall in August, September and October is 12 mm, 6 mm and 2 mm. Write the expression for the combined average per day. 5

(ii) Evaluate it for 31, 30 and 31 days.

END OF QUESTION PAPER — **Total: 40 Marks**